

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

Semester: January 2015 - April 2015

Max. Marks: 100**Duration: 3 hrs****End Semester Exam**

Class: S.E

Name of the Course: Theory of Computer Science

Semester: IV

Course Code: UCEC405

Branch: CMPN

Instructions:

- (1) All Questions are Compulsory
- (2) Figures to the right indicate full marks
- (3) Make suitable assumptions wherever necessary with proper justification.

Question No.		Max. Marks
Q1	Answer any five questions:	25
(a)	Give statement for pumping Lemma for context free languages and mention steps to prove them.	
(b)	Differentiate between NFA and DFA	
(c)	What are recursive and recursively enumerable languages?	
(d)	Explain the necessity of Myhill Nerode Theorem to prove a language to be regular	
(e)	Define with examples Moore and Mealy Machines	
(f)	Give examples of PDA as a decider and acceptor of CFG	
(g)	Construct an NFA with epsilon transition for the following RE: $(00+11)^*(10)^*$	
Q2		
(a)	Design a PDA that checks for well formed parenthesis	10
	OR	
	Design a PDA corresponding to the grammar $S \rightarrow aSa bSb \epsilon$	10
(b)	Design a Turing Machine to recognize a language $L = \{a^n b^n a^n n \geq 1\}$	10
(c)	Discuss the subset and superset relation between Regular Languages, Context Free Languages and Context Sensitive Languages	05
Q3		
(a)	Design a Finite State Machine to check whether given decimal number is divisible by '5'.	10
(b)	Design a Turing Machine for the following language,	10
	OR	

Q4		
(a)	For the following given grammar, find the left most derivation, right most derivation and parse tree for the string "001222" $S \rightarrow 0S \mid 1A \mid 2B \mid \epsilon$ $A \rightarrow 1A \mid 2B \mid \epsilon$ $B \rightarrow 2B \mid \epsilon$	10
(b)	Using the rules for conversion, convert the following CFG to CNF $S \rightarrow bA \mid aB$ $A \rightarrow bAA \mid aS \mid a$ $B \rightarrow aBB \mid bS \mid b$ OR	10
(b)	What is the Halting Problem? Prove that it is undecidable.	10
(c)	Give the closure properties for Regular Languages.	05